

5CM510 – Network System development

Lecture 14: VANETS and UAV Networks

Lecture Outline

- Vehicular Communication and VANETs
- Information dissemination in Vehicular Networks
- Broadcasting and routing In Vehicular Networks
- Unmanned Aerial Vehicle (UAVs) Networks

Section 1

Vehicular Ad Hoc Networks (VANETs) and Communications

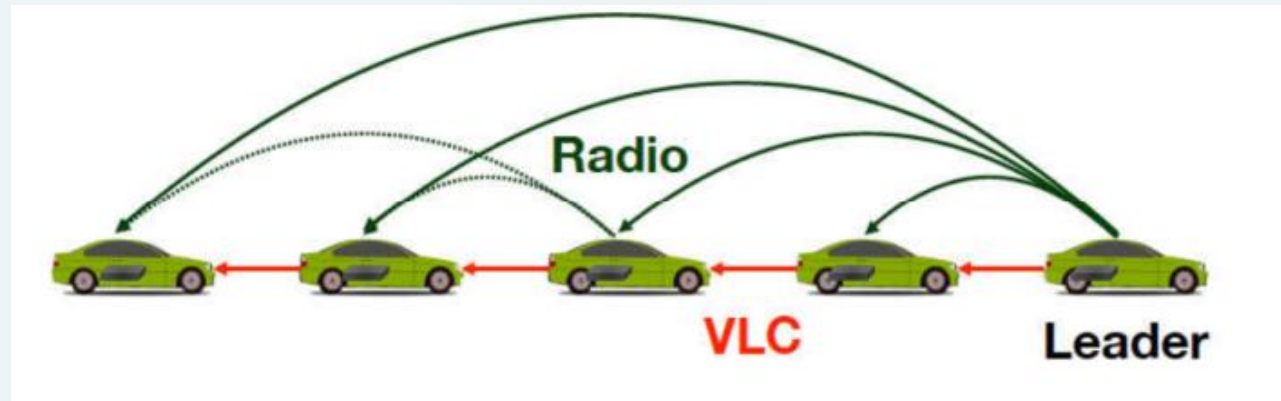
Vehicular Networks (and VANETS)

- The environment of vehicular networks is developing very fast. Their application to real life is getting closer and will become a reality; autonomous vehicles will need to interconnect to speak to each other and make joint decisions.
- Vehicular ad hoc networks (VANETs) operate with little or no permanent infrastructure and are characterized by:
 - High mobility
 - Fixed road networks (Roadside units or RSUs)
 - Predictable speed and traffic patterns in congested conditions
 - Very few power constraints or storage limitations
- Unlike other communication systems, in which the primary goal is to achieve high message throughput, VANETs aim primarily for communication reliability and fast dissemination.

Vehicular Communications

- Importance :traffic management, Road safety, and smart mobility.
- Communication pathways include:
- Vehicle-to-vehicle (V2V): messages are transmitted between neighbouring vehicles. This includes “one-hop” and “multi-hop” messaging scenarios in which vehicles communicate directly with other vehicles or through intermediary vehicles.
- Vehicle-to-infrastructure (V2I): messages are transmitted between vehicles and roadside units located on nearby arterial road intersections or highway on-ramps.
- Vehicle-to-pedestrian (V2P): messages are transmitted between vehicles and pedestrians who send and receive messages via their phones or other wireless devices.
- Vehicle-to-everything (V2X) mode includes two communication modes: mode 3 and mode 4.
 - Mode 3: the cellular network allocates and manages radio resources used by vehicles for their communication. Signalling transits through the eNodeB or gNB (Base station) or the MEC datacentre which virtualizes the eNodeB.
 - mode 4: vehicles select radio resources autonomously to perform direct V2V communications, which means directly from one vehicle to another.

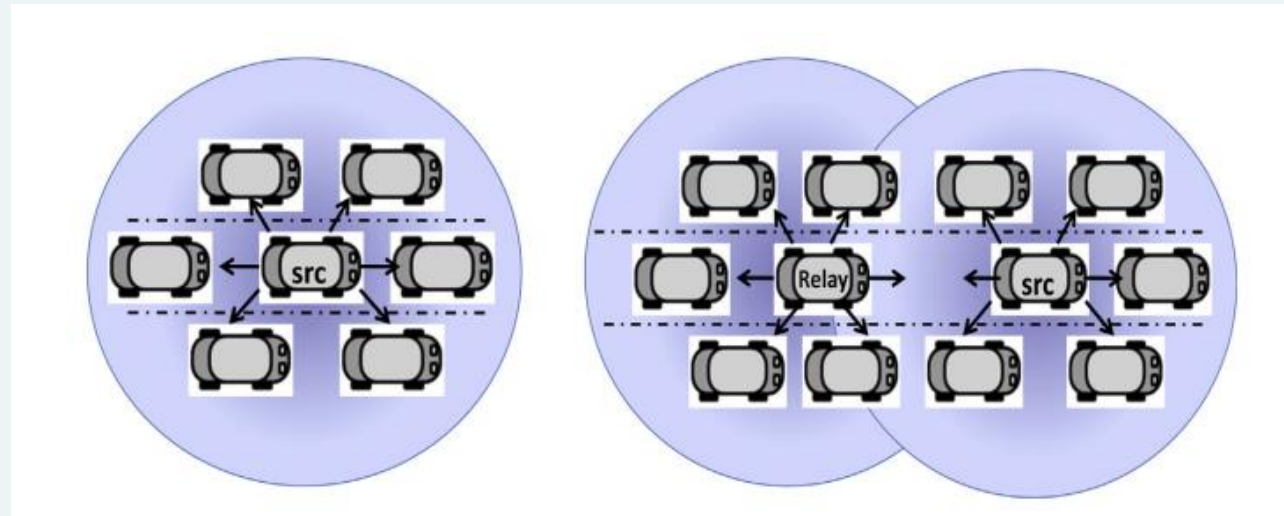
V2V using Visible light communication



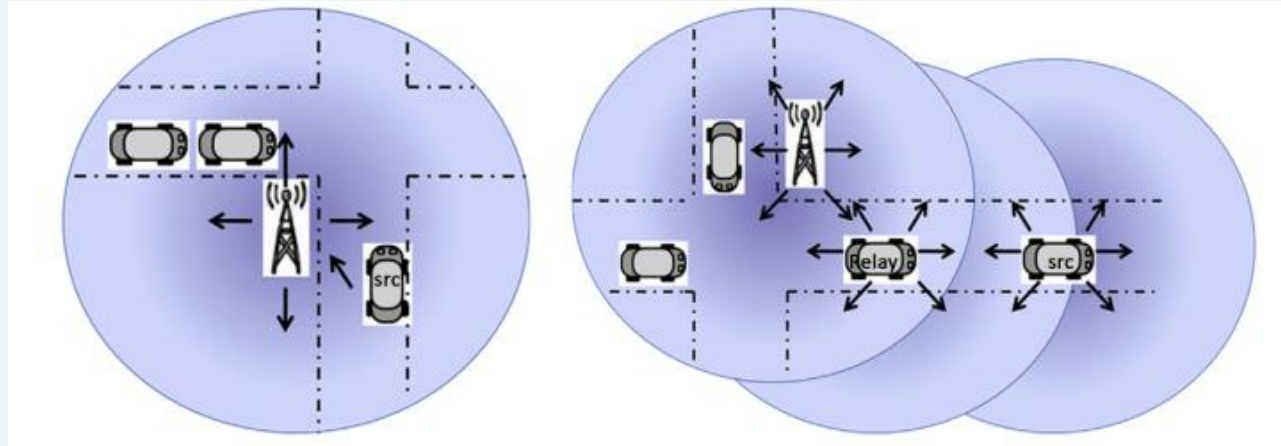
Vehicular Communications: Applications

- These vehicular communication networks can help improve safety, the environment, and mobility. For example, V2V and V2I systems use information on acceleration and braking behaviors of nearby vehicles to track dangers beyond a driver's line of sight, helping to prevent collisions.
- With vehicles in communication with each other, "platooning" is possible, which reduces vehicle headways and aerodynamic drag, helping to improve fuel efficiency.
- V2P systems improve the safety of pedestrians crossing at intersections and facilitate carpooling and ridesharing, by providing people real-time information about their rides.
- Additionally, the network can also provide useful information, such as route guidance, and even entertainment to passengers and drivers.

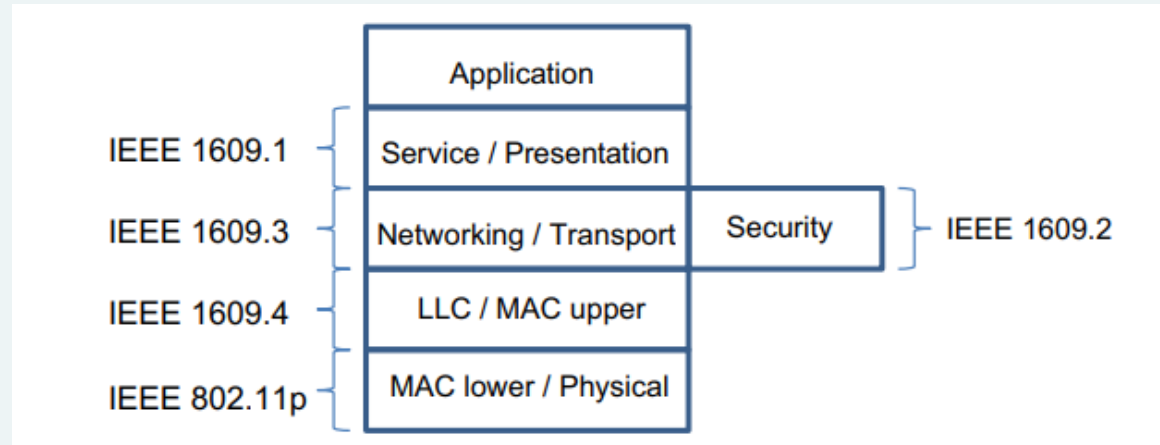
Safety: Single hop V2V and Multi-hop V2V



Safety: Single hop I2V and Multi-hop I2V including V2V



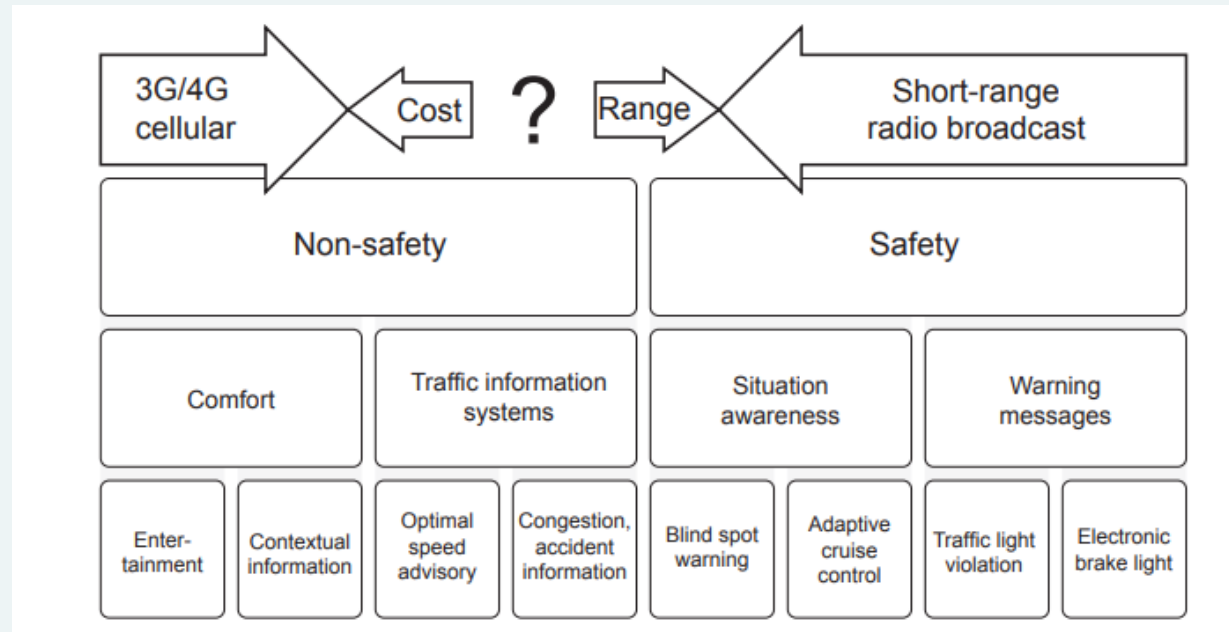
Enabling Technologies: OSI model and Dedicated short range communication(DSRC) Standards



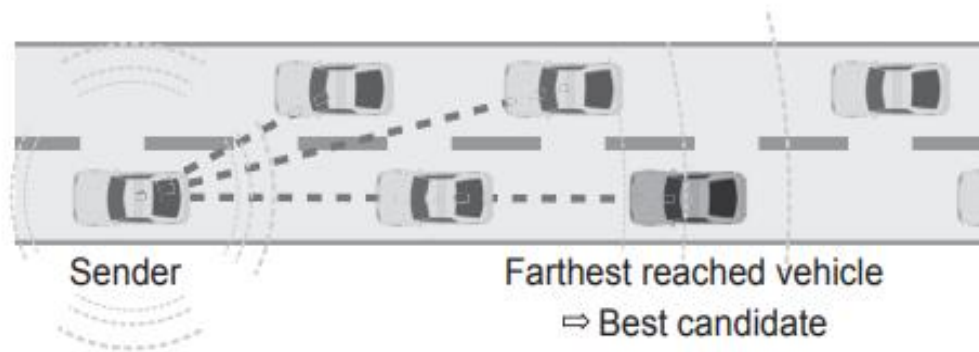
Section 2

Information Dissemination in Vehicular Networks

Taxonomy of Inter Vehicular Communication application and technologies

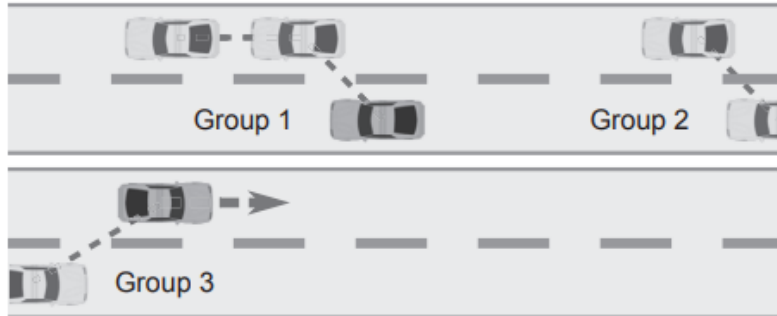


Intelligent flooding and geocasting



- The general idea of broadcast suppression is to select best candidates among all vehicles that might be able to forward the message.
- Broadcast suppression using slotted 1-persistence - greedily make progress in all directions by preferring the vehicles farthest from the source node.
- All other nodes overhearing the forwarded message are supposed to cancel (to suppress) their rebroadcast.
- To solve the broadcast storm problem more effectively, given that some data, e.g., notification about a critical accident or very generic parking place information, is only relevant to cars following or those driving to a certain place – this concept is called geocasting.

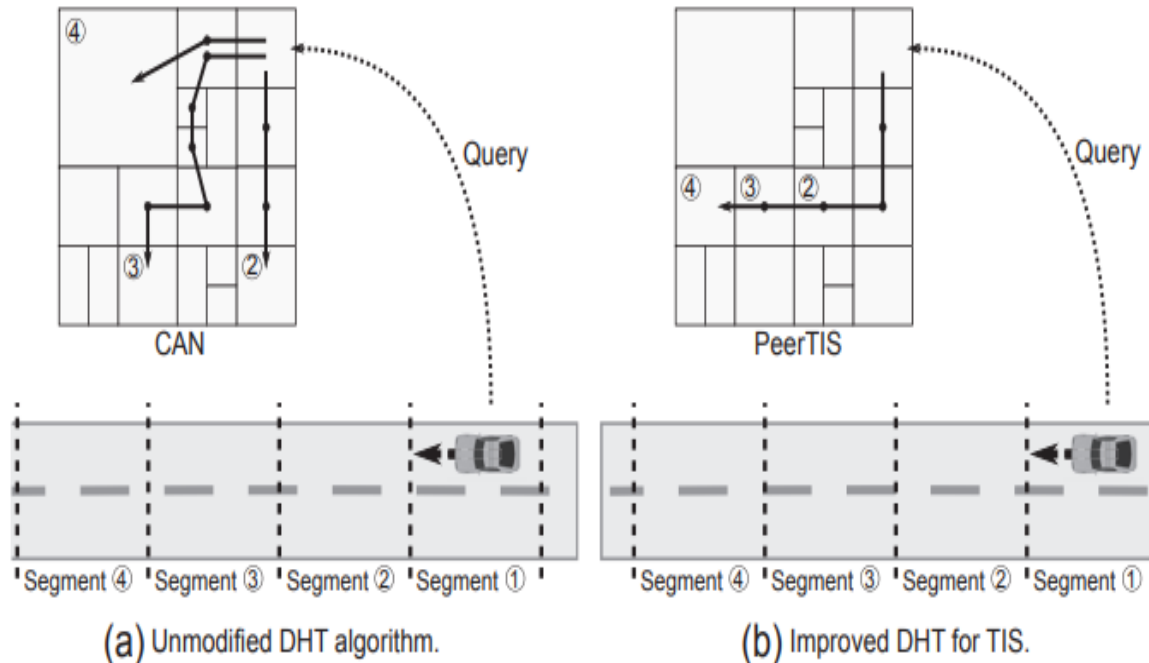
Store-carry-forward



- The store carry-forward communication principle exploits spontaneous contacts between mobile systems to disseminate information in an epidemic manner.
- As can be seen, three groups (or clusters) of cars are driving on a freeway.
- Let us assume that the leader of group 1 starts to disseminate a message. Groups 1 and 2 are disconnected.
- The only possible communication alternative, and the one chosen by distributed vehicular broadcast (DV-CAST), is to use vehicles in group 3 to carry information from group 1 to group 2. This way, communication becomes more efficient in terms of coverage, but, of course, transmission latencies become unpredictable.

Using cell-networks: Peer-to-Peer traffic information system

- PeerTIS's motivation is that traffic information systems need to keep data current with high spatial resolution.
- The concept is based on that of distributed hash tables (DHTs). Internet-based DHTs such as Chord and Content Addressable Networks (CANs) provide efficient distributed storage over many nodes in IP-based networks.
- unmodified DHT algorithm - the employed hashing may lead to random distribution of data causing long query times and high load in the network for subsequent queries.
- PeerTIS uses CAN, but replaces the hashing algorithm with neighborhood information according to the underlying map.



Section 3

Broadcast protocols and routing in Vehicular Networks

Broadcasting in Vehicular networks

- Broadcasting in vehicular networks is more prevalent than other transmission methods because of need to facilitate rapid, brief communications with nearby vehicles.
- However, the success of these message dissemination schemes relies on our ability to develop and implement routing protocols to specifically address the low latency requirements in a highly mobile environment.
- These latency requirements arise from the need to disseminate safety related information in as few as 20 ms and also because transmissions made to fast-moving vehicles are typically only relevant at an instant.

Broadcast Protocols

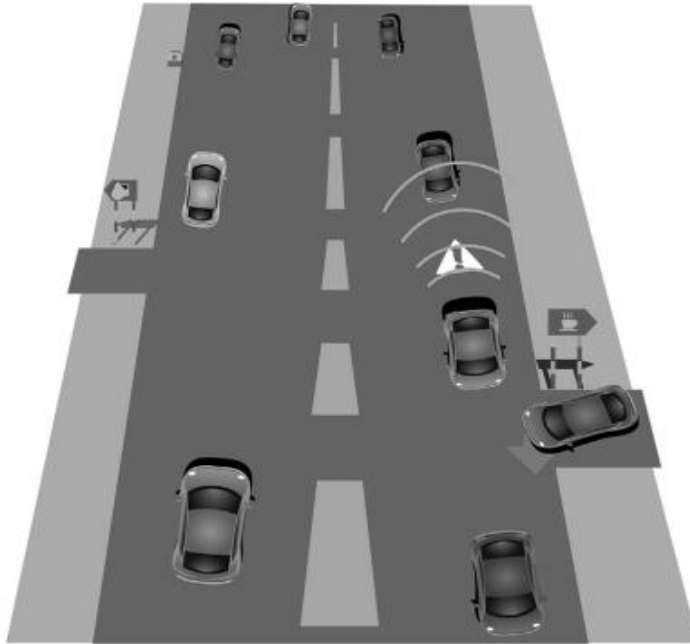
Protocol, year introduced	Communication characteristics					Traffic characteristics		
	Location/ Position-based	Distance/ Hop-based	Cluster- based	Probabilistic- based	Network simulator	Traffic- based	Highways	Arterial roads/ Local streets
UMB, 2004	✓	✓			WS			✓
Traffic View, 2004					ns-2	✓	✓	
MDDV, 2004	✓				QualNet		✓	✓
ODAM, 2004	✓	✓			ns-2			
OAPB/DB, 2005	✓			✓	ns-2			✓
AMB, 2006	✓				WS			✓
SB, 2006	✓	✓						
MHVB, 2006	✓				ns-2		✓	
D-FPAV, 2006		✓			ns-2		✓	
TRRS, 2007		✓					✓	
REACT, 2007	✓				ns-2	✓	✓	
DV-CAST, 2007				✓	ns-2		✓	
FB, 2007		✓						

- Urban multi-hop broadcast (UMB) and ad hoc and multi-hop broadcast (AMB): select vehicles farthest away from the information source as relay nodes.
- Smart broadcast (SB): use a dynamic backoff timer for media access control (MAC) contention window adjustment to improve the efficiency of packet transmission.
- Multi-hop vehicular broadcast (MHVB): adjusts the packet transmission interval with position-based method.

Vehicular network routing

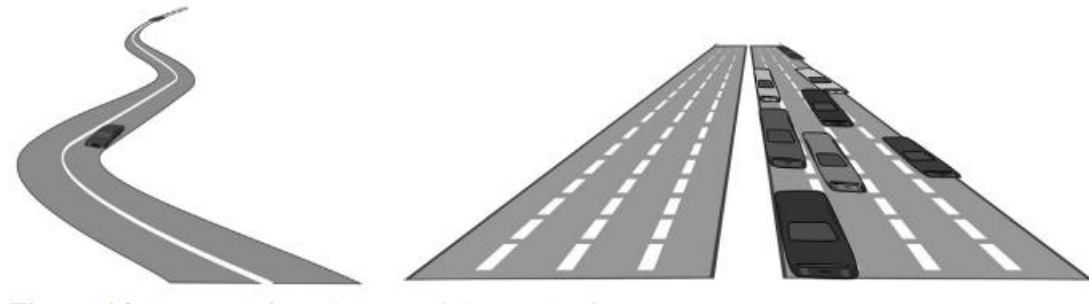
- Vehicular communication systems are created with many applications in mind (Krishnan, Bai, & Holland, 2010), but probably the main motivation is safety, along with traffic management and several commercial uses.
- Applications can be characterised and classified according to different criteria, but two parameters that are especially relevant (from a routing point of view) are:
- the region of interest for the application (that is, the geographical region where the information is significant for the different entities that participate in the application)
- the duration of the event.

Vehicular routing applications



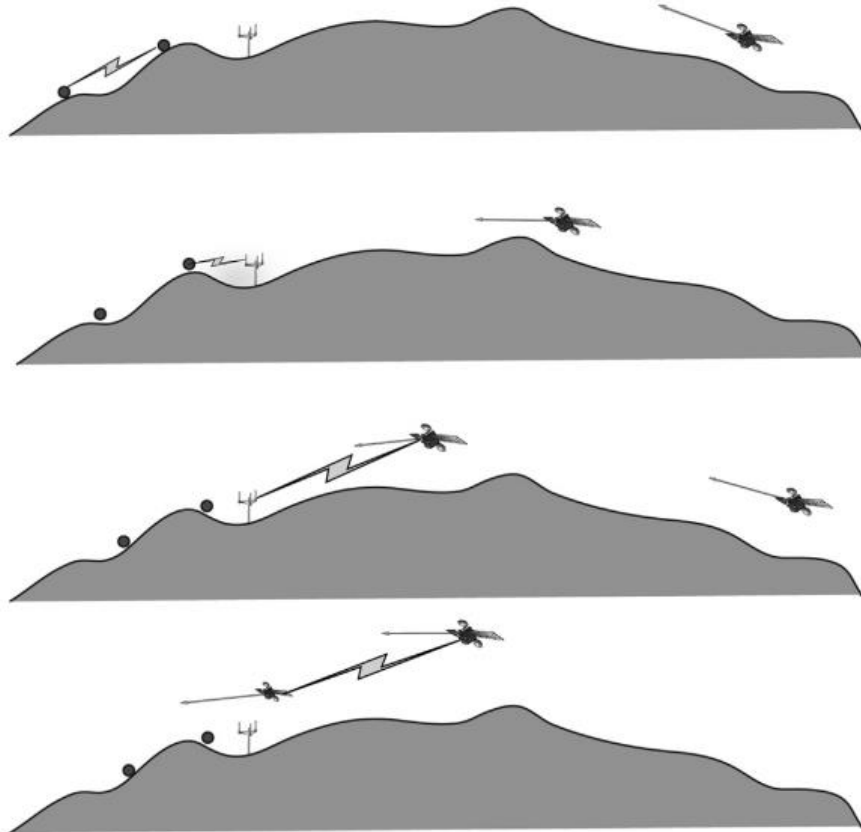
- For example, a hard or panic braking event from a vehicle should be notified only to near vehicles that are just a few hundred metres behind.
- The notification should not be sent to vehicles that are kilometres away.
- This notification is valid only until the brake is released, and the vehicle is back at normal speed.

Sparse and Dense topologies in a highway



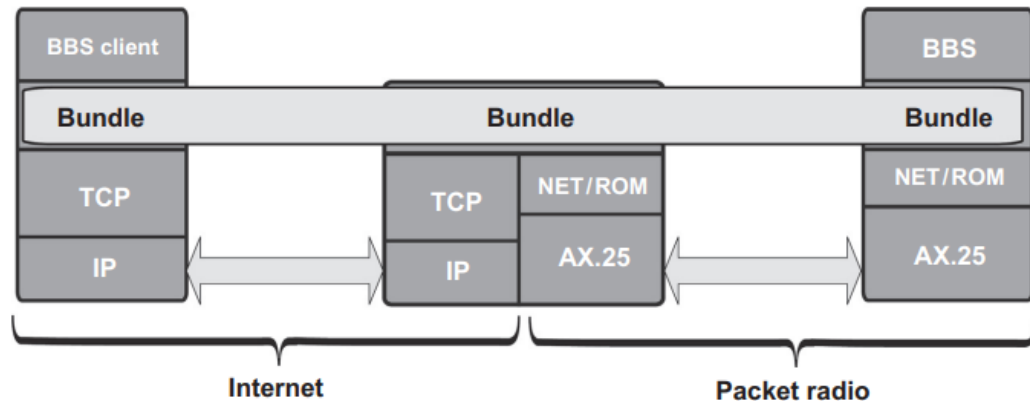
- There are two main types of network topologies in a highway.
- highly dense networks during rush hours,
- sparsely populated networks during late night hours (this is also true for freeways in remote areas or lightly used roads).
- consider a highly dense topology, flooding is the natural choice to extend the broadcast model over a multi-hop environment.
- Packets should include enough information to allow receivers rebroadcast it.
- That information may include the direction of the vehicle, the expected coverage area (expressed in distance or in time), etc.

Store and forward strategy



- The Internet Research Task Force (IRTF) created the Delay Tolerant Networking Research Group (DTNRG).
- The aim was to address the architectural and protocol design principles to provide interoperable communications with and among extreme and performance-challenged environments where continuous end-to-end connectivity cannot be assumed (<http://irtf.org/dtnrg>).

Delay tolerant networking stack and overlay network



- Delay-tolerant networks (DTNs) are designed to tolerate connection disruption, and to achieve this feature,
- The designers decided to create the 'bundle layer': an overlay layer that uses persistent storage to avoid the loss of data during the disconnections.
- The 'Bundle Protocol' is the base to create the bundle layer. It can run over IP or any other protocol specially adapted to any particular environment (sensor networks, long distance communication, etc.).
- This protocol collects all the information the different applications want to transmit in a bundle, in a manner that resembles e-mail.

Section 4

UAV Networks

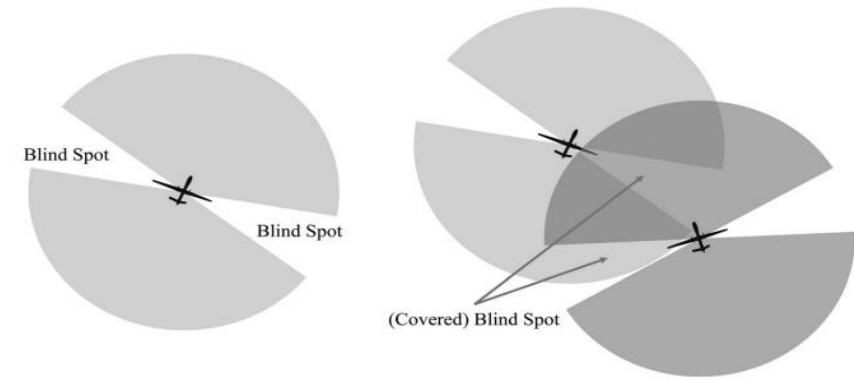
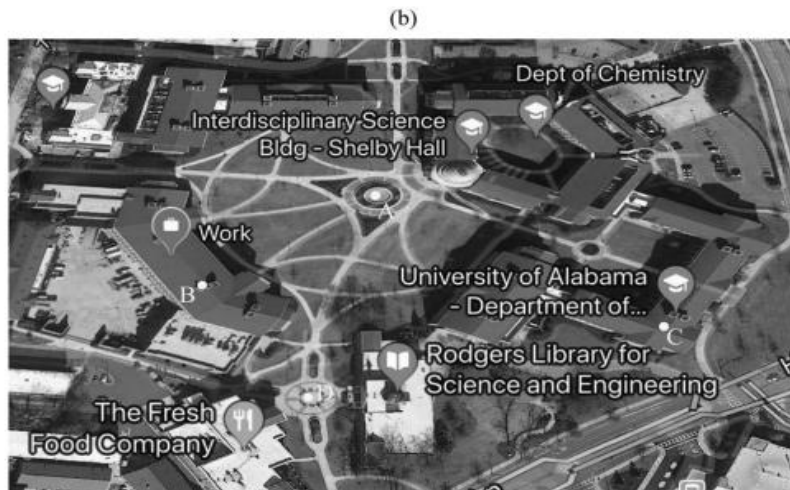
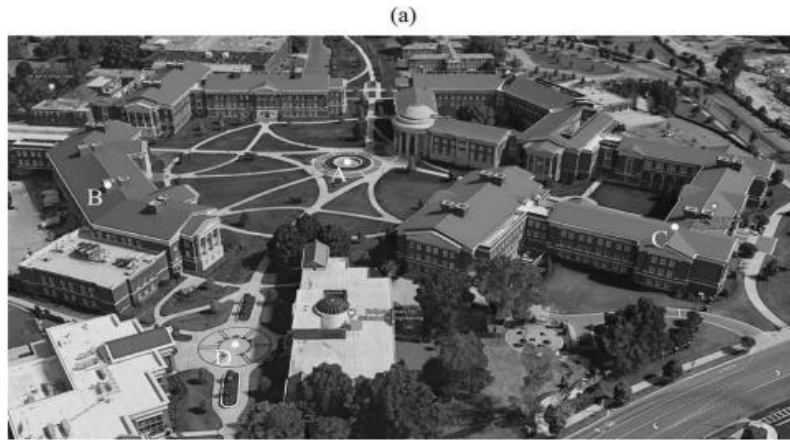
Unmanned Aerial Vehicles (UAV) Networks

- Unmanned aerial vehicles (UAVs) are aircraft without the on-board presence of a pilot or crew. Any unmanned aircraft must be deemed to have a pilot.
- The aircraft may be controlled remotely, or it may be running autonomously with a program loaded on the aircraft's on-board computer, but the person who loaded the program and initiated the running of the autopilot is still regarded as the pilot.
- To prevent the ill-utilization of UAVs, more and more nations have issued rules and regulations on the design, manufacture, distribution and setting up no-fly zones against UAVs for privacy purposes

Issues with the single UAV system

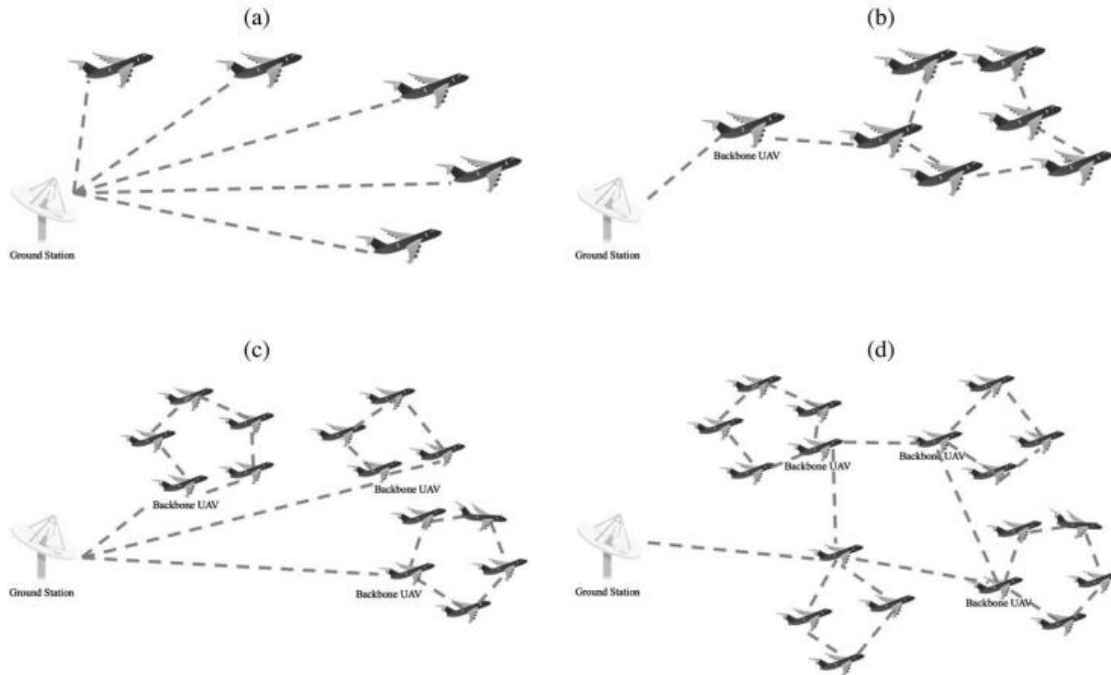
- The most common flying strategy of UAVs is the single-UAV system.
- The UAV is set off individually, controlled and communicated by a human through a single-phase, two-way channel.
- The single-UAV (agent) is relatively short ranged, once the command has been issued by the controller, the drone executes and gives feedback via a wireless network.
- This end-to-end communication method has three possible design flaws:
- The communication quality is dependent on the travel distance of the drone, where the increased distance of the UAV yields a poorer connection.
- The short-range response time limits this drone from tasks that require a relatively long distance.
- The drone is not able to respond to the change of aerial environment intelligently and give feedback in time, so that if something happened to the drone that terminated the commanding channel, the UAV-at-large might not be able to retract back to the user.

Single UAV Vs Multi-UAV Systems



- The search range of single-UAV forms a 2D geographical map with relative coordinates x , y . Due to the limitation of its design, it is possible to leave a traceable blind spot on the map.
- A multi-UAV utilizes a group of small UAVs working simultaneously on a mission such that the field of view (FOV) formed has a depth of zero or a very limited blind spot.

Communication Architectures between UAVs and ground station



- Four different architectures for communication and networking for network drones, in which UAVs can be automatically or remotely grounded by ground crews:
- (a) centralized UAV network,
- (b) UAV ad hoc network,
- (c) multi-group UAV network
- (d) multi-layer UAV ad hoc network.

Communication Architectures

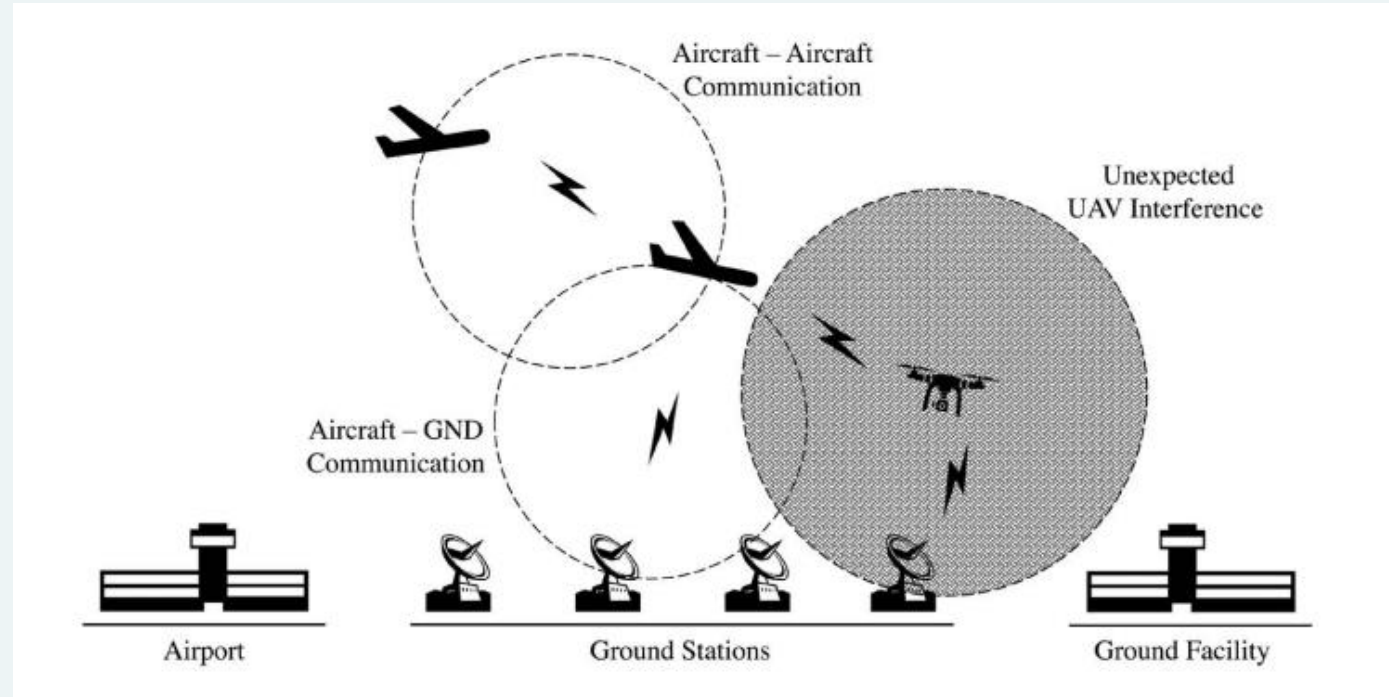
Communication Architectures between UAVs and Ground Stations

Communication Type	Connection Architecture	Pros	Cons
Centralized	Centralized connection	Direct communication with ground Overall supervising	Limited distance No connection between UAVs Blind spots exist
Decentralized	UAV ad hoc	Interconnection between UAVs Coverage area extended	Limited distance Single loop failure
Multi-group		Direct communication with ground Blind spots killed Different flight and communication types	Semi-centralized Lack robustness
Multi-layer ad hoc		Freely route between UAVs Reduced computation load Reduced communication load	Backbone UAV interchangeable No single point failure

Typical aircraft control under the ad hoc networking protocol with unexpected UAV signal invasion

- Aircraft commonly utilize ad hoc networks. If there are no shielding techniques between UAVs and the aircraft-ground system, the UAVs are strictly banned near certain locations.
- Typically, the connections are set up among ground facilities (e.g., airport), ground stations and in-air aircraft.
- However, illegally flying UAVs also sends out a signal under the same networking protocols. This unexpected signal, if not well shielded, might cause normal communication failure or even aircraft accidents.
- Under this circumstance, more than 150 nations all over the world issued detailed regulations specifying UAV no-fly zones (or no-drone zone according to the FAA) for safety and privacy concerns.

Aircraft control with unexpected UAV interference



Summary

- The advent of autonomous vehicles, while likely further away than the 2020 date that several manufacturers are proposing, appears unstoppable; semiautonomous vehicles, which will be on the road much sooner than fully autonomous ones, will also require successful implementation of these technologies.
- Perfecting communication in VANETs and overcoming the liability and public agency jurisdictional issues preventing large-scale adoption will be absolutely necessary to the safe and efficient movement of those vehicles.
- Successful broadcast routing protocols are designed to address specialized vehicular network layouts (highway, arterial, or urban roads) and specific safety scenarios.
- The DTN paradigm was specially designed for these kinds of networks: the ‘challenged networks’, that is, networks that do not satisfy the requirements of TCP/IP protocols to provide end-to-end connectivity.
- we can imagine a scenario where a Mars rover collects information about the planet surface with its sensors. It creates a bundle with the measurements but has to wait until an orbiting satellite is on line-of-sight to transmit it.
- UAV networks have emerged as a promising technique to rapidly provide wireless coverage over a geographical area.
- There are many advantages of multi-UAV systems: (1) economy, (2) flexibility, (3) continuity, (4) speediness, (5) higher accuracy, (6) sustainability, and (7) ease of problem solving. Nonetheless, despite the listed advantages, the most challenging design issue deals with communication.



UNIVERSITY OF
DERBY

University of Derby, Kedleston Road, Derby, DE22 1GB

T +44 (0)1332 591044 E opendays@derby.ac.uk



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derby.ac.uk